

Hammad Ahmad

AI / Machine Learning Engineer | LLMs · RAG · Knowledge Graphs

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [ORCID](#)

Right to work: UK Graduate visa to December 2027. Eligible to work now, no sponsorship required.

PROFILE

AI/ML engineer with an MSc (Merit) in Applied AI and a first-author Springer publication. I build retrieval-augmented pipelines that pair vector search with knowledge graphs so answers stay grounded and citable: on FinLaw-UK, every citation is resolved against a Neo4j graph before it reaches the user, and the system refuses rather than answering when retrieval is weak. I take systems from retrieval design through backend APIs to production, including a voice-agent backend whose call latency I cut 54% (2.4s to 1.1s).

TECHNICAL SKILLS

Machine Learning, NLP & LLMs : PyTorch, scikit-learn, XGBoost, LightGBM, Retrieval-Augmented Generation (RAG), Sentence Transformers, vector embeddings, semantic search, cross-encoder re-ranking, RAGAS evaluation, Ollama

Engineering & Data : Python, TypeScript, JavaScript, SQL, FastAPI, Flask, REST APIs, React, Next.js, PostgreSQL, Neo4j, pandas, NumPy

MLOps & DevOps : MLflow, Docker, Git, GitHub Actions (CI/CD), Linux

EXPERIENCE

Jobzyl (self-employed)

May 2026 - Present

Founder & Sole Engineer, [jobzyl.com](#)

Remote

- Built and operate a live job-aggregation platform serving a **617K-posting index**: cache-first reads with live parallel fan-out to 20 third-party sources on miss, per-source timeout budgets and failure isolation so a single slow board never blocks the response, and SSE streaming (first results ~1.4s).
- Ingestion, expiry and staleness handling across sources; server-side deduplication, salary normalisation to annual, a scheduled alerts pipeline, and an ATS-style CV matcher with parsing kept browser-side so CV text never leaves the device. Next.js, TypeScript, Python, FastAPI; CV text encrypted at rest with hard-delete flows.

Quality Control Assistant, Myton Food Group (Morrisons), Bradford, UK

Apr 2026 - Present

Outlyst

Oct 2025 - Mar 2026

AI / Machine Learning Engineer (fixed-term contract)

Leeds, UK (Remote)

- Built and deployed an AI voice-agent system for outbound calling (Retell AI, FastAPI): backend dialogue-flow logic, automated call flows and lead-qualification rules, handling 2,100+ calls and cutting manual calling workload by roughly 50%.
- Reduced average call latency 54%, from 2.4s to 1.1s, by profiling backend async I/O and restructuring connection pooling; built an internal micro-CRM for lead tracking that removed external CRM licensing costs.

University of Bradford

Jan 2025 - Sep 2025

MSc Research Project, Machine Learning & LLMs

Bradford, UK

- Built **FinLaw-UK** (code + benchmark), a graph-augmented RAG system for UK financial regulation: hybrid BM25 and BGE-small dense retrieval fused by reciprocal rank fusion, cross-encoder re-ranking, 2-hop Neo4j traversal, and a verifier resolving every citation against the graph before it reaches the user. Mistral 7B via Ollama, fully local.
- Re-measured the submitted evaluation and found the reported source-accuracy and citation-quality figures were regex shape-checks, not correctness measures: the scorer awarded a flat 0.85 to any citation-shaped string, and 103 of 110 rows carried that constant. Published the correction and the true graph-verified rate of 3 in 110, which motivated the citation normaliser, re-ranker and refusal gate.

COMSATS University Islamabad

Jul 2023 - Jul 2024

Research Intern, Data Science

Islamabad, Pakistan

- Benchmarked 11 classifiers for diabetes risk on a 253,680-record CDC BRFSS dataset (14% positive class), comparing random over-sampling, SMOTE and ADASYN with resampling confined to training folds; Random Forest performed best on ROC-AUC and sensitivity. Deployed it as a public risk-screening web app (React, Flask): [DiabetesSense](#).

EDUCATION

University of Bradford

Sep 2024 - Sep 2025

MSc, Applied Artificial Intelligence and Data Analytics (Merit)

Bradford, UK

- Dissertation**: FinLaw-UK, a graph-augmented retrieval system for UK financial regulation.

COMSATS University Islamabad

Sep 2020 - Jul 2024

BS, Bioinformatics

Islamabad, Pakistan

PUBLICATIONS

Ahmad, H., Khan, M.U., & Azam, M. (2024). Comparative Analysis of Machine Learning Methods for Enhancing Sleep Efficiency and Prediction. *ICSMAI 2024* (Springer). DOI: [10.1007/978-3-031-66854-8_1](#)

LANGUAGES: English (IELTS 7.0, CEFR C1) | Urdu (Native) | German (A1.2)